References and findings - literature review

Sl	Author	Link	Abstract
1	**Sardari et. al**	https://doi.org/10.1016/j.est.2025.119782	The widespread adoption of lithium-ion batteries (LIBs) in electric vehicles, portable electronics, and renewable energy systems has intensified the demand for effective thermal management strategies to ensure safety, performance, and longevity. Among emerging solutions, phase change materials (PCMs) have attracted significant interest for their passive thermal regulation capabilities, particularly their high latent heat and isothermal behavior. However, the research landscape is fragmented, with varied materials, configurations, and integration strategies explored across the literature. This review aims to provide a comprehensive synthesis of recent advances in PCM-based battery thermal management systems (BTMS), highlighting their thermal performance, structural innovations, and hybrid configurations with active cooling methods. Drawing on experimental, numerical, and theoretical studies, the review identifies key design parameters affecting efficiency, including PCM composition, nano-enhancements, encapsulation techniques, and system architecture. The review examines emerging research trends such as the use of machine learning for BTMS optimization, development of multifunctional PCMs with flame-retardant and structural properties, and cost-benefit analyses supporting commercial viability. Despite notable progress, challenges remain in material stability,

			scalability, and integration complexity. This review concludes by outlining future research directions, including the standardization of performance metrics, exploration of bio-based PCMs, and adaptive control strategies.
2	Abanoz and Arici	https://doi.org/10.1016/j.est.2025.118799	Electric vehicles (EVs), which are critical in controlling carbon emissions and protecting ecological harmony and human well-being, are reinforcing their key role in the transportation sector. However, EV batteries can ensure high performance, prolonged service life, and safe driving only with proper and effective battery thermal management systems (BTMSs). The focus and major contribution of this study to literature is to reveal the role of phase change material's (PCM's) latent heat to the thermal performance of batteries, differentiating the latent heat from the overall thermal energy storage capacity which includes both sensible and latent heat. For that purpose, the phase stable material approach, which is a material with the identical thermophysical properties as PCM but does not undergo phase transformation, was applied. In the study, aluminum radial fin intensified approaches are utilized to enhance the thermal performance of PCM-based BTMS. Additionally to the reference model adding only PCM, 3 different simulation scenarios were analyzed: i) PCM + 1 fin (Design-I), ii) PCM + 3 fins (Design-II), iii) PCM + 5 fins (Design-III). The study was conducted under 5C discharge conditions by integrating 3 different PCMs (RT31, RT38, and RT42) and

			the impacts of fin distribution, fin dimensions, and housing diameters on thermal behavior of PCM and maximum temperature of battery were numerically studied. The results revealed that the fins enhanced the low thermal conductivity of the PCM by increasing the heat transfer area and can be further improved depending on the intensification of the fins. The refinement of the fin distribution accelerated heat transfer. Compared to the reference case, the temperature decrease was approximately 23.22 %, 12.64 %, and 11.12 % for RT31, RT38, and RT42 in Design-III, respectively. The increase in fin length led to a temperature reduction of up to 2.23 °C, demonstrating greater effectiveness compared to the increase in fin thickness. Reducing the inner diameter of the housing from 34 mm to 26 mm slightly increased the temperature values, but the influence on the liquid fraction was considerable. At the end of the discharge process, the liquid fraction reached from 36 % to 89 %, 28 % to 76 % and 19 % to 65 % for RT31, RT38, and RT42, respectively. Contribution of latent heat in reducing the temperature was 21.94 % in the reference case, 23.51 % in Design-I, 25.4 % in Design-II, and 33.62 % in Design-III.
3	Possidente et. al	https://doi.org/10.1016/j.firesaf.2023.103991	Equations are given in the Eurocode standard EN 1993-1-2 on the estimation of temperature in insulated steel cross sections when assuming uniform steel section temperature, sometimes called a lumped heat capacity assumption. When deriving these equations the

			temperature of the exposed insulation surface is assumed equal the surrounding fire gas temperature. That is a simplification that may provide inaccurate results for heavily insulated steel sections, when the heat capacity of the insulation is considerably higher than that of the steel section. Therefore, a new lumped mass formulation suited for heavily insulated steel sections is proposed in this paper. It accounts for the heat transfer resistance between the fire and the insulation surface as well as for the heat capacity of the insulation. The accuracy of the predictions made with the proposed new formulation are compared with results of accurate 1-D finite element numerical analyses considering insulation materials with several combinations of thermal properties and thicknesses. Analyses with the proposed formulation is shown to yield accurate and generally on the safe side estimations of steel temperature in comparison to the finite element (FE) calculations. The new formulation is applicable to lightly as well as heavily insulated steel sections.
4	Li et. al	https://doi.org/10.1016/j.applthermaleng.2024.122794	As an important intermediary between the green energy and human society, the lithium-ion battery has promising prospects in the new energy vehicles, energy storage, and green development fields. However, lithium-ion batteries can generate a large amount of heat during operation. In addition, excess temperature or big temperature difference of the surface of the lithium-ion battery can reduce its performance, which also results in

			reducing its operational life and even triggers incidents such as fires and explosions. Therefore, studying efficient thermal management methods for lithium-ion batteries is crucial for increasing their application range. This paper proposes a novel thermal management method by combining 3-D finned tubes with PCM. The thermal management system maintains a battery operating temperature less than 45 °C and a maximum temperature difference less than 2.4 °C. The influence of different fin structure parameters on the melting dynamics of PCM and the heat transfer performance is then discussed. The obtained results show that under each solid–liquid ratio of PCM, the heat transfer performance increases with the increases of the fin height, while it initially decreases and then increases with the increase of the width and axial spacing. Moreover, the heat transfer performance of the 3-D finned tube with different PCM liquid–solid ratios is investigated. The obtained results demonstrate that for constant fin parameters, the heat transfer reaches its highest performance for a PCM liquid–solid ratio greater than 2:2. The findings of this study lay a foundation for the design of efficient thermal management systems incorporating PCM applications.
5	Saxxon A. et al	https://doi.org/10.3390/batteries10040136	Battery design efforts often prioritize enhancing the energy density of the active materials and their utilization. However, optimizing thermal management systems at both the cell and pack levels is also key to achieving mission-relevant battery design. Battery thermal management systems, responsible for managing the thermal profile of battery cells, are crucial for balancing the trade-offs between battery performance and

			lifetime. Designing such systems requires accounting for the multitude of heat sources within battery cells and packs. This paper provides a summary of heat generation characterizations observed in several commercial Li-ion battery cells using isothermal battery calorimetry. The primary focus is on assessing the impact of temperatures, C-rates, and formation cycles. Moreover, a module-level characterization demonstrated the significant additional heat generated by module interconnects. Characterizing heat signatures at each level helps inform manufacturing at the design, production, and characterization phases that might otherwise go unaccounted for at the full pack level. Further testing of a 5 kWh battery pack revealed that a considerable temperature non-uniformity may arise due to inefficient cooling arrangements. To mitigate this type of challenge, a combined thermal characterization and multi-domain modeling approach is proposed, offering a solution without the need for constructing a costly module prototype
6	Barış Kavasoğulları	https://doi.org/10.34248/bsengineering.1545174	The design and numerical analysis of the two-layer PCM (Phase Change Material)-based thermal management system for a 18650-type lithium-ion battery have been performed. In relation to simulation, the coefficient of thermal conductivity and melting temperature of the first layer of PCMs are varied. Other parameters are made identical to that of the next layer's parameters in order that the generation of two different layers of PCMs can be attained: PCM-1 and PCM-2. To obtain a more realistic approach in the numerical analysis, the battery thermal model was created in the COMSOL-MATLAB interface using the experimental internal resistance data obtained for 18650 type Li-ion batteries in the literature. While a cheaper and more accessible material with a thermal conductivity of 0.2 W/mK and a melting point of

			50 °C was used in the PCM-2 layer, the thermal conductivity was changed as 0.2, 1 and 5 W/mK and the melting point was changed as 30, 40 and 50 °C in the PCM-1 layer. In this way, for PCM layers with different thickness (tpcm), the system was optimized at two different discharge rates, 5C and 7C. As a result of the numerical analysis, it was determined that the optimum tpcm, kpcm,1 and Tm values for the 5C discharge rate were 2 mm, 0.2 W/mK and 40 °C, respectively; and the optimum tpcm, kpcm,1 and Tm values for the 7C discharge rate were 4 mm, 5 W/mK and 40 °C, respectively.